Parth Karnawat

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Experience

MUFG (Mitsubishi UFJ Financial Group) **Dec 2025 – Present**
Senior Analyst, Model Risk Management *Bengaluru, India*

- **Initial Margin and PFE Models** | *Counterparty Credit Risk*
  - (Ongoing) Baseline validation of IM, and PFE models of MUFG Securities Canada, for TRS on bonds and equities.
- **AI Automation** | *Model Validation Workflows*
  - (1) Built a hybrid-RAG over internal documents, code, procedures and policies; (2) a custom MCP tool to integrate model inventory (GRC Archer); (3) and an AI agent for automated validation report drafting.
  - Promoted adoption of GitHub Copilot, Git, and LaTeX for model validation workflows, developed a LaTeX validation report template and conducted team-wide technical training on Git, Neural Networks/LLMs, and AI tools.
- **Scorecard: Non-Project Power & Utilities** | *Credit Rating Model*
  - Periodic re-validation of a Tier 1 rating model. Conducted a comprehensive review of the model, in line with SR 11-7 guidelines, and provided recommendations for model improvement and risk mitigation.

Standard Chartered Bank **Jan 2023 – Nov 2025**
Model Development, Credit Risk – IFRS9 and Stress Testing Models *Bengaluru, India*

Associate *Mar 2025 – Nov 2025*

- **Bahrain Personal Loans** | *PD, EAD, and LGD Models*
  - Developed a full suite of IFRS9-compliant PD, EAD, and LGD models to estimate ECL for the BAU scenario.
  - First suite of models by the team to be developed entirely in Python.

Analyst *Jan 2023 – Mar 2025*

- **SAS to Python Migration**
  - Led the migration of model development workflow from SAS to Python, by building an internal Python library to (1) efficiently convert large SAS datasets to Parquet using SAS APIs and `pyarrow`, (2) implemented internal methodologies for PD, EAD, and LGD models.
  - Employed `Dask` and `DuckDB` for processing larger-than-memory datasets and leveraging distributed computing.
- **Korea Personal Loans** | *PD Model*
  - Developed a segmented, two-stage (logistic–linear) regression model for Probability of Default (PD) estimation, incorporating statistical insights and macroeconomic factors to support Expected Credit Loss (ECL) forecasting under business-as-usual and stressed scenarios, in alignment with IFRS9 requirements and business objectives.
  - Redevelopment of models led to a reduction of ECL by ∼USD 8M, from USD 40M to USD 32M.
- **Stakeholder Management**
  - Coordinated with validation, implementation, and monitoring teams to address queries, secure model approvals, and ensure timely deployment and performance tracking.
  - Provided valuable portfolio insights to Korea and Bahrain teams to support data-driven risk policy decisions.

Education

Indian Statistical Institute **July 2020 – June 2022**
Master of Mathematics (M.Math) *Kolkata, India*

- Grade: 74.55%, First Division [Transcript | Syllabi]
- Thesis: *"Rates of Convergence in Urn Models"* [Report] Supervisors: Prof. Arup Bose and Prof. Krishanu Maulik

Indian Statistical Institute **July 2017 – June 2020**
Bachelor of Mathematics (B.Math [Hons.]) *Bengaluru, India*

- Grade: 85.25%, First Division with Distinction [Transcript | Syllabi]
- CS II Project: *"'Engineers' Quest To Fast Approximate Computations"* [Report]

World Quant University **Oct 2024 – July 2027** (expected)
Diploma in Financial Engineering *Online, Distance Learning*

- Grade: 87% [Transcript]
- Relevant Coursework: Financial Markets, Financial Data, Time Series Analysis, Derivative Pricing, Stochastic Modeling, Machine Learning in Finance, Deep Learning for Finance

Technical Skills

Advanced: Python, SAS, SQL
Basic: C++, HTML/CSS, JavaScript, Java, R
AI/ML: pandas/numpy/scipy, scikit-learn/statsmodels, Dask/PySpark, DSPy, Weaviate
Tools: Git, LaTeX, bash/Linux, SageMath

Relevant Coursework

Probability Theory: Measure Theory , Stochastic Processes , Martingales , Brownian Motion , Weak Convergence And Empirical Processes , Ergodic Theory
Statistics: Statistical Inference(Hypothesis Testing, Point and Interval Estimation) , Linear/Logistic Regression , Time Series Analysis , Bayesian Statistics
Applied Mathematics: Numerical Analysis , Data Structures and Algorithms , Optimization , Differential Equations and PDE's , Graph Theory , Advanced Linear Algebra

Certifications

GitHub Copilot - [Link]	June 2025
Foundations in Financial Engineering (WQU) - [Link]	Feb 2025
Advanced SAS - [Link]	March 2024
Base SAS - [Link]	May 2023

Teaching / Leadership / Projects

Monsoon Math Camp Instructor | *Math Education, Markov Chains* **July 2020**

- Taught at Monsoon Math Camp, a free immersive math camp aimed at introducing advanced high school students in India to topics in advanced mathematics. [Website]
- Co-instructed a course, which exposed students to Markov Chains and Random Walks.

Limit 2019 – Online Math Olympiad | *Web Development, ISI Bangalore* **November 2019**

- Led frontend and backend development for Limit 2019, an online math Olympiad hosted by ISI Bangalore. [Website]
- Designed, maintained and hosted(on AWS) the infrastructure enabling seamless participation, submission, and evaluation of the Olympiad, with almost 10k participants.

Undergraduate Research Article | *Resonance Journal of Science Education* **May 2018**

- Authored a published article in Resonance Journal of Science Education titled *"Points at Which Continuous Functions Have the Same Height"*. [Link]
- Explored a real analysis challenge problem from undergraduate coursework through formal mathematical exposition.